

SPIKING NEURAL NETWORKS FOR MULTIMODAL NEUROIMAGING

An Analysis of Current Trends and the NeuCube Brain-Inspired
Architecture

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October 21, 2025

Introduction

- AI has revolutionized neuroimaging for disease detection and multimodal data integration.
- ANNs face challenges with temporal dynamics and computational cost.
- SNNs mimic biological neurons using discrete spikes.
- NeuCube is a specialized SNN framework for spatiotemporal brain data.

- ① **Improving NeuCube (2023):** Enhanced EEG pattern recognition via transfer learning.
- ② **ANN-SNN Conversion (2023):** Retains ANN accuracy with SNN efficiency.
- ③ **BackEISNN (2022):** Balanced excitatory-inhibitory neurons for adaptive learning.
- ④ **SpikingJelly (2023):** Python-based framework for spike-based intelligence.

Fundamentals of SNNs and Neuromorphic Computing

Neuromorphic Computing: Brain-inspired computing: High energy efficiency via parallel, event-driven processing.

Spiking Neural Networks (SNNs): Use discrete, neuron-like spikes to process dynamic, time-varying data.

Key Concepts:

- **Spikes:**
- **Membrane Potential:**
- **Temporal Coding:**
- **Synaptic Plasticity:**
- **Refractory Period:**

- **Leaky Integrate-and-Fire (LIF):**
 - **Integrate:** Input spikes increase membrane potential.
 - **Leak:** Adds biological realism; potential decays over time.
 - **Fire:** Emits a spike when potential reaches threshold.
 - **Reset:** Potential resets after firing.

SNN Neuron Models

fig3.jpg

- **Integrate-and-Fire (IF):** The simplest model that integrates incoming current.
- **Izhikevich:** Balances biological accuracy with computational efficiency.
- **Hodgkin-Huxley:** The most detailed and accurate model, but computationally heavy.
- **AdEx:** An advanced model that mimics neuron fatigue.

- **Unsupervised** : based on the relative timing of pre- and post-synaptic spikes.
 - 1 **Spike-Timing Dependent Plasticity (STDP)**

If a presynaptic neuron fires **before** a postsynaptic neuron → **strengthen** the synapse (Long-Term Potentiation).

If it fires **after** → **weaken** the synapse (Long-Term Depression).
 - 2 **Hebbian Learning**

“Neurons that fire together, wire together.”

- **Supervised Learning using Surrogate Gradients**

- Surrogate derivatives are used to approximate gradients.
- Enables backpropagation-like training for SNNs.
- Trained on labeled datasets like ANNs.

- **Reinforcement (R-STDP):**

- Synaptic weights based on reward signals.
- Combines spike activity with feedback from the environment.
- Useful for robotics or neuromorphic control tasks.

SNNs encode information in spikes, mainly in two ways:

- **Rate coding:** Number of spikes in a time window.
Example: More spikes $\rightarrow$ higher intensity.
- **Temporal coding:** Exact timing of spikes.
Example: The first neuron to fire represents the strongest input.

Spike Encoding

fig4.jpg

SNNs vs ANNs

image.png

- **Structural Neuroimaging (MRI):**
- **Functional Neuroimaging (EEG/fMRI): Applications:** Brain-Computer Interfaces (BCIs), disease diagnosis (e.g., Alzheimer's, epilepsy), and classifying cognitive states.
- **Multimodal Data Integration:**

Development Challenges: Adoption is hindered by a lack of pre-built models, specialized software, and standardized formats.

Software Simulators:

- snnTorch
- SpikingJelly
- Nengo DL
- NeuCube

Hardware Platforms:

- **Intel Loihi2** :A research chip with programmable neuron models and on-chip learning. 1.15 billion neuron system based on this architecture;95% accuracy in gesture recognition with 200x lower power than a GPU.
- **IBM NorthPole**: A chip focused on high-performance, energy-efficient model inference, rather than SNN development.
- **BrainChip Akida**: Paired with a CPU for hybrid execution.includes a CNN-to-SNN conversion tool. The platform has shown 98% accuracy recognizing fast-moving objects on edge devices.
- **Lynxi KA200**: A high-performance AI accelerator designed for edge computing and autonomous driving.

NeuCube: Architecture & Implementation

- Brain-inspired 3D structure for spatiotemporal mapping.
- Integrates multimodal data.
- Supports hybrid learning (unsupervised + supervised).

Architectural Components:

- **Input Data Stream:**
- **Gene Regulatory Network (GRN):**
- **Neurogenetic Cube:**
- **Output Module:**

The NeuCube Framework

fig2.jpg

Implementation & Flexibility:

- NeuCube supports flexible implementation on conventional computers, GPUs, and neuromorphic hardware (e.g., Loihi and SpiNNaker).
- Integrates diverse data, adapt its structure, and incorporate GRNs makes it a versatile hybrid platform.

Case Study 1: MRI Classification

- Compared SNN (SpikingJelly) and CNN (AlexNet) for Alzheimer's detection.
- CNN: 85.9% accuracy vs SNN: 78% (faster but less precise).
- SNNs less effective for static MRI data.

fig 1.jpg

Case Study 2: EEG and fMRI Analysis

- NeuCube achieved 97% accuracy in EEG-based disease detection.
- fMRI: 90% accuracy in cognitive state classification.
- Outperformed traditional ML models for temporal data.

Case Study 3: Longitudinal MRI Data

- NeuCube trained on 6-year MRI dataset for dementia prediction.
- Predicted disease 2 years in advance with 91% accuracy.
- Demonstrates potential for long-term predictive neuroimaging.

Hybrid Systems: Integrating ANNs and SNNs

- Combines CNN's spatial analysis with SNN's temporal efficiency.
- Used in neuroprosthetics and adaptive neuro-fuzzy systems.
- Enhances brain-computer interface performance.

Challenges and Future Outlook

- Lack of standardization and benchmarking.
- Sensitive to noise in EEG data.
- Difficult training due to non-differentiable spikes.
- Ethical issues: privacy, bias, and clinical safety.
- **Future:** predictive modeling, personalized medicine, and BCIs.

Research will focus on predictive modeling for early diagnosis, personalized medicine, and advanced Brain-Computer Interfaces.

Conclusion

- SNNs outperform ANNs in temporal neuroimaging tasks like EEG/fMRI.
- NeuCube demonstrates strong multimodal integration and adaptability.
- Hybrid ANN-SNN models show real-world potential.
- Standardization and clinical validation are key future steps.

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